

Comprehensive Examination Strategy Guide

RGPV BT-105 Engineering Graphics

PYQ Analysis & Preparation Module

First Year B.Tech (All Branches)

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1 Examination Structural Dynamics and Analytical Review

An exhaustive evaluation of the Rajiv Gandhi Proudhyogiki Vishwavidyalaya (RGPV) B.Tech first-year Engineering Graphics (BT-105) examination reveals a highly consistent, structured evaluation matrix. This assessment framework is meticulously designed to test spatial reasoning, mathematical precision, geometrical construction capabilities, and modern technological proficiency in computer-aided design.

The standard RGPV BT-105 examination paper is designed for a three-hour duration, presenting a maximum score of 70 marks. The paper consistently features eight distinct questions. Examinees are mandated to attempt any five questions, granting them a strategic choice in navigating the examination. Consequently, each question carries an equal weight of 14 marks. The internal architecture of these questions typically features a bifurcation into part (a) and part (b), carrying 7 marks each, ensuring that students are tested on multiple distinct concepts within a single question.

1.1 Typology of Examination Questions

The evaluation metrics encompass a hybrid of manual drafting skills, mathematical calculations, theoretical knowledge, and Computer-Aided Design (CAD) proficiency:

- **Numerical and Drafting Problems (60% - 70%):** These problems demand precise geometrical construction on drawing sheets. Examples include calculating the Representative Fraction (RF) to construct diagonal or vernier scales, projecting complex solids that are inclined to multiple reference planes, and generating isometric views from orthographic projections.
- **Theoretical Questions (20% - 30%):** These assess the student's foundational understanding of engineering graphics principles. Frequent topics include the differentiation between First Angle and Third Angle Projection systems.
- **CAD & Digital Drafting (10% - 20%):** Almost every paper includes questions related to CAD command sequences, editing commands, the layering concept, and BIM (Building Information Modeling).

1.2 Difficulty Level Trends

A chronological assessment highlights a distinct evolution in the difficulty matrix. Earlier papers (2019-2020) focused heavily on fundamental manual constructions, simple plain scales, and elementary solid projections. Recent papers (2023-2025) indicate a pronounced shift toward combined complexities and modern industrial relevance. Solids are frequently modeled in complex orientations requiring multi-stage auxiliary plane projections. The CAD segment has transitioned from basic command definitions to application-based scenarios.

2 Exhaustive Syllabus Breakdown and Priority Matrix

Based on historical cyclicity, the syllabus can be categorized into distinct units with a specific probability of appearance.

Unit / Chapter	Specific Topics & Sub-topics	Priority	Probability
1. Scales & Curves	Diagonal/Vernier Scales, RF calculation	High	95%
	Conic Sections (Ellipse, Parabola)	High	85%
	Special Curves (Epicycloid, Involute)	Medium	65%
2. Orthographic Projections	Lines inclined to both HP and VP	High	95%
	First Angle vs. Third Angle Theory Projections of Planes (Polygonal)	High Medium	85% 70%
3. Projection of Solids	Solids inclined to one, parallel to other	High	90%
	Solids resting on generator or base edge	High	85%
4. Sections & Dev.	True shape of sections	High	90%
	Lateral development of truncated solid	High	85%
5. Isometric Views	Isometric Scale vs. True Scale	High	85%
	Isometric view of combined solids	High	90%
6. CAD & BIM	Edit Commands (Move, Copy, Mirror)	High	95%
	Layering Concept & BIM	High	90%

3 High-Frequency Questions and Thematic Repeats

Mastering these specific recurring questions provides a massive strategic advantage:

1. **The CAD Layering Concept:** "Explain the Layering concept with examples. How is it implemented in CAD software?" (Appears in ~75% of papers).
2. **First Angle vs. Third Angle Projection:** "Differentiate between First Angle and Third Angle Projections."
3. **Building Information Modeling (BIM):** "What is BIM? How does a BIM 'wall' differ from traditional 2D CAD?"
4. **Area-based RF Diagonal Scale:** "An area of X sq cm on a map represents Y sq km on the field. Find the RF and draw a diagonal scale."
5. **CAD Edit Commands:** "Name and explain five edit/modify commands used in CAD (e.g., Move, Copy, Rotate)."

4 Predicted Questions for the Upcoming Examination

Q1. Scales: Construct a Vernier scale to read meters, decimeters, and centimeters, measuring up to 6 meters, where 1 meter is represented by 2.5 cm. Find RF and mark 4.38 meters.

Q2. Curves: Construct an epicycloid given that the rolling circle diameter is 50 mm and directing circle diameter is 150 mm. Draw a tangent and normal.

Q3. Lines: A line segment AB, 80 mm long, has end A 15 mm above HP and 20 mm in front of VP. Inclined at 30° to HP and 45° to VP. Draw projections and locate traces.

Q4. Planes: A regular pentagonal lamina of 30 mm side rests with one base edge on HP. The highest point is 25 mm above HP. The resting edge is inclined at 30° to VP.

Q5. Section & Development: A right circular cylinder (50 mm dia, 70 mm height) rests vertically. Intersected by a section plane perpendicular to VP and inclined at 45° to HP, bisecting the axis. Draw true shape and lateral development.

Q6. Isometric: A sphere of 40 mm diameter is placed centrally on top of a square prism (base 50 mm, height 25 mm). Draw the composite isometric view.

Q7. CAD Theory: Define BIM and explain its lifecycle management utility. Detail CAD modify commands.

5 Full-Length Model Question Paper

Time: Three Hours

Maximum Marks: 70

Instructions: Attempt any five questions out of eight. All questions carry equal marks (14 Marks each). Assume missing dimensions suitably.

Q.1 (a) An area of 144 cm^2 on a map represents an actual geographic area of 36 km^2 . Calculate the Representative Fraction (RF). Construct a diagonal scale to show kilometers, hectometers, and decameters, long enough to measure up to 10 km. Indicate a distance of 7 km, 5 hectometers, and 6 decameters. **(7 Marks)**

Q.1 (b) Construct a continuous conic section curve (ellipse) using the general eccentricity method, where the distance of the focus from the directrix is 50 mm and eccentricity is $2/3$. Draw a tangent and normal. **(7 Marks)**

Q.2 (a) Differentiate conceptually, graphically, and theoretically between First Angle Projection and Third Angle Projection. **(7 Marks)**

Q.2 (b) A straight line PQ, 75 mm long, has its end P situated 20 mm above HP and 15 mm in front of VP. The true length is inclined at 45° to HP and 30° to VP. Draw orthographic projections and locate its traces. **(7 Marks)**

Q.3 (a) A regular hexagonal lamina of 30 mm side rests on one of its edges on HP. The surface is inclined at 45° to HP, and the resting edge makes 30° with VP. Draw final projections. **(7 Marks)**

Q.3 (b) A right circular cone (base 50 mm dia, axis 70 mm long) is lying on one of its extreme generators on the HP. Its axis remains parallel to the VP. Draw top and front projections. **(7 Marks)**

Q.4 A right regular hexagonal prism, base edge 25 mm and axis 60 mm, rests on its base on HP with two opposite faces parallel to VP. A cutting plane, perpendicular to VP and inclined at 45° to HP, bisects its vertical axis. Draw the sectional top view, true shape, and exact geometric development of the lateral surface. **(14 Marks)**

Q.5 (a) Detail the core principles of Isometric Projection. Illustrate Isometric Scale construction. **(7 Marks)**

Q.5 (b) A cube (40 mm edge) is placed centrally on top of a cylindrical block (60 mm dia, 25 mm height). Draw the isometric view. **(7 Marks)**

Q.6 (a) Define Building Information Modeling (BIM) and how it transcends 2D CAD drafting. **(7 Marks)**

Q.6 (b) Describe the standard workflow for drafting a single BHK floor plan using CAD software. **(7 Marks)**

Q.7 (a) What is the concept of "Layering" in CAD? Provide an architectural example. **(7 Marks)**

Q.7 (b) Write the syntax and functional descriptions of: MOVE, COPY, ROTATE, and MIRROR. **(7 Marks)**

Q.8 Write short notes on any three: (i) Epicycloid (ii) Viewports (iii) Wireframe vs Solid Modeling (iv) Vernier scale principle. **(14 Marks)**

6 Comprehensive Step-by-Step Solved Answers

Solution 1(a): Diagonal Scale (Area-Based RF)

Step 1: Derivation of Representative Fraction (RF)

The RF is the ratio of drawing size to actual size. For area dimensions, use the square root:

$$RF = \sqrt{\frac{\text{Area on map}}{\text{Actual Area}}} = \sqrt{\frac{144 \text{ cm}^2}{36 \text{ km}^2}} = \frac{12 \text{ cm}}{6 \text{ km}}$$

Convert kilometers to centimeters (1 km = 100,000 cm):

$$RF = \frac{12}{6 \times 100,000} = \frac{12}{600,000} = \frac{1}{50,000}$$

Step 2: Calculation of Length of Scale (LOS)

$$LOS = RF \times \text{Maximum distance} = \left(\frac{1}{50,000}\right) \times 10 \text{ km}$$

Convert maximum distance to cm:

$$LOS = \left(\frac{1}{50,000}\right) \times (1,000,000 \text{ cm}) = 20 \text{ cm}$$

Drafting Steps: Draw a 20 cm line. Divide into 10 primary blocks (1 km each). Subdivide the first block into 10 vertical sections (0.1 km / 1 hm each). Create 10 horizontal strips across the height (0.01 km / 1 dam each). Draw diagonal lines in the first block. To mark 7.56 km, start at primary 7, move to secondary 5, go up the diagonal to horizontal strip 6.

Solution 1(b): Ellipse via Eccentricity Method

Given eccentricity $e = 2/3$. Since $e < 1$, it is an ellipse.

1. Draw Directrix (DD'). Mark point C on it. Draw horizontal axis perpendicular to DD' through C .
2. Mark Focus (F) 50 mm from C on the axis.
3. Divide CF into $2 + 3 = 5$ equal parts (10 mm each). Locate Vertex (V) such that $VF = 20$ mm and $VC = 30$ mm.
4. Draw vertical line from V . Mark E on it where $VE = VF = 20$ mm. Draw a line from C through E .
5. Mark points 1, 2, 3 on the axis, draw verticals to intersect extended CE line at $1', 2', 3'$.
6. With F as center and radius $1 - 1'$, draw arcs on the vertical line passing through 1 to get points P_1, P_1' . Repeat for all points and trace the smooth curve.

Solution 2(a): First Angle vs Third Angle Projection

- **Object Placement:** First Angle assumes the object is in the 1st Quadrant (above HP, in front of VP). Third Angle assumes the 3rd Quadrant (below HP, behind VP).
- **Sequence:** First Angle is Observer $\rightarrow$ Object $\rightarrow$ Plane. Third Angle is Observer $\rightarrow$ Plane (transparent) $\rightarrow$ Object.
- **Views:** In First Angle, Top View is drawn *below* the Front View. In Third Angle, Top View is drawn *above* the Front View.

Solution 2(b): Projections of a Line Inclined to Both Planes

Using the Rotation Method:

1. Draw XY line. Mark p' 20 mm above XY and p 15 mm below XY. Draw horizontal loci through them.
2. Draw True Length (75 mm) from p' at 45° to get q'_1 . Draw locus of q' . Drop vertical to get q_1 on the locus of p .
3. Draw True Length (75 mm) from p at 30° to get q_2 . Draw locus of q . Raise vertical to get q'_2 on the locus of p' .
4. Rotate $p - q_1$ up to the locus of q to get final Top View pq . Rotate $p' - q'_2$ down to locus of q' to get final Front View $p'q'$.

Solution 6(a): Building Information Modeling (BIM)

BIM transcends standard 2D CAD by using dynamic 3D intelligent data models rather than static vector lines. A "wall" in standard CAD is just parallel lines with X, Y coordinates. In BIM, a wall object holds physical metadata: material density, thermal resistance, cost, and load-bearing capacity. Changing an object in a BIM 3D model automatically updates all related 2D floor plans, sections, and material bills simultaneously, eliminating discrepancies.

Solution 7(b): CAD Modification Commands

- **MOVE:** Select object, specify base point, specify displacement. Translates objects without altering geometry.
- **COPY:** Select object, specify base point, specify displacement. Leaves the original intact while creating a duplicate.
- **ROTATE:** Select object, specify base pivot point, input rotation angle (+ for counterclockwise).
- **MIRROR:** Select object, specify two points for the mirror line. Creates a mathematically symmetrical replica, halving drafting time for symmetrical parts.